1. EC2 Basics

Amazon Elastic Compute Cloud (Amazon EC2) provides scalable computing capacity in the AWS Cloud. It eliminates the need to invest in hardware up front, so you can develop and deploy applications faster.

- **AMI (Amazon Machine Image):**
  - A template that contains the software configuration (OS, application server, and applications) required to launch your instance.
  - **Region Scoped:** AMIs are locked to a specific region. To use an AMI in another region, you must **copy** it.
  - **Public vs. Private:** You can create your own AMIs and choose to keep them private or share them publicly.
- **User Data:**
  - Script used to run commands when an instance **first launches**.
  - Used to bootstrap instances (e.g., install updates, install software, download files from S3).
  - Runs with `root` privileges.
- **Instance Metadata:**
  - Data *about* your instance (e.g., private IP, public IP, hostname, IAM role name).
  - Accessible from within the instance via URL: `http://169.254.169.254/latest/meta-data/`
  - *Exam Tip:* Do not confuse **User Data** (boot script) with **Metadata** (instance info).

2. EC2 Instance Types Basics

Understanding instance families is crucial for selecting the right resource for a workload.

Naming Convention: e.g., `m5.2xlarge`

- **m**: Instance Class (Family)
- **5**: Generation (Higher is newer)
- **2xlarge**: Size (CPU/RAM capacity)

Key Instance Families (Mnemonic: "FIGHT DR MC. P.X.")

Family Code	Type	Use Case
G, P, Tr, Inf	**Accelerated Computing**	Graphics, Graphics Processing, Floating Point Number calculations, Machine Learning, AI.

Family Code	Type	Use Case
I, D, H	**Storage Optimized**	High IOPS, Data Warehousing, Distributed File Systems. **I** instances are great for databases needing high I/O (NoSQL).
M	**General Purpose**	Balanced Compute/Memory/Networking. Good for application servers, gaming servers, small DBs.
C	**Compute Optimized**	High performance processors. Batch processing, media transcoding, scientific modeling, dedicated gaming servers.
R, X, Z	**Memory Optimized**	High RAM. In-memory databases (Redis, Memcached), Real-time processing of big data (Apache Spark).
T	**Burstable**	General Purpose but with the ability to "burst" CPU. Uses **CPU Credits**. Good for workloads with idle time (e.g., dev environments).

- *Exam Tip:* If the scenario mentions "High Performance Computing (HPC)" or "Batch Processing," think **C** family. If it mentions "In-memory database" or "Cache," think **R** family.

3. Security Groups & Classic Ports Overview

A **Security Group** acts as a **virtual firewall** for your EC2 instances to control incoming and outgoing traffic.

- **Scope:** Acting at the **Instance level**, not the subnet level.
- **Stateful:** If you allow traffic IN, the response traffic is automatically allowed OUT (and vice versa).
- **Rules:**
  - Can only **ALLOW** rules (cannot create DENY rules).
  - Implicitly **DENIES ALL** inbound traffic by default.
  - Allows **ALL OUTBOUND** traffic by default.
- **Referencing:** You can authorize a specific Security Group ID instead of a CIDR block. This is best practice for multi-tier apps (e.g., Allow DB SG to accept traffic only from WebApp SG).

Common Ports to Know

Port	Protocol	Service
22	SSH	Linux Login (Secure Shell)
3389	RDP	Windows Login (Remote Desktop)
80	HTTP	Unsecured Web Traffic

Port	Protocol	Service
443	HTTPS	Secured Web Traffic
21	FTP	File Transfer
3306	TCP	MySQL / Aurora / MariaDB
5432	TCP	PostgreSQL

- *Exam Tip:* If you cannot connect to your instance, it is almost always a **Security Group** issue (time out). If you get a "Connection Refused" error, the Security Group is likely fine, but the application on the instance isn't running.

4. SSH Overview

SSH (Secure Shell) is used to command a Linux instance remotely.

- **Key Pairs:** AWS uses public-key cryptography.
  - **Public Key:** Stored by AWS on the EC2 instance.
  - **Private Key:** Stored by you (the `.pem` file). **If you lose this, you cannot recover it.**
- **Putty (Windows < 10):** Requires converting `.pem` to `.ppk` using PuttyGen.
- **Chmod 400:** On Mac/Linux, you must secure your key file permissions before connecting: `chmod 400 my-key-pair.pem`.
- **EC2 Instance Connect:** A browser-based SSH connection that works without managing key files manually (uses IAM permissions).

5. EC2 Instance Purchasing Options

Choosing the right pricing model is a common cost-optimization question on the exam.

Option	Description	Best For
On-Demand	Pay by the second (Linux/Win) or hour. No commitment, highest cost.	Short-term, spiky, or unpredictable workloads. First-time apps.
Reserved (Standard)	1 or 3-year commitment. Up to 72% discount. Can modify AZ, scope, networking type.	Steady-state usage (e.g., core database).
Reserved (Convertible)	1 or 3-year commitment. Up to 54% discount. Can change instance **family**, OS, tenancy, and payment option.	Steady-state usage where software needs might change.

Option	Description	Best For
Savings Plans	Commitment to a specific usage (e.g., $10/hour) for 1 or 3 years. **EC2 Instance Savings Plan** (specific family/region) or **Compute Savings Plan** (flexible across EC2, Fargate, Lambda).	Most flexible long-term option. Compute Savings Plans are very popular for exam answers regarding flexibility + cost.
Spot Instances	Use spare capacity. Up to 90% discount. Can be interrupted with 2 minutes notice.	Stateless, fault-tolerant, flexible workloads (Batch jobs, Image processing, CI/CD).
Dedicated Host	Physical server fully dedicated to your use. Visibility into sockets/cores.	Compliance requirements (BYOL - Bring Your Own License) that charge per socket/core.
Dedicated Instance	Instances run on hardware dedicated to you (single tenant), but you don't control placement.	Compliance requirements needing single-tenancy but NOT specific hardware control.
Capacity Reservation	Reserves capacity in a specific AZ. **No billing discount** (charged On-Demand rate whether you run it or not).	Ensuring you can launch an instance in a specific AZ during a critical event (e.g., Black Friday).

6. Spot Instances & Spot Fleet

Spot instances are the cheapest option but come with the risk of termination.

- **Spot Price:** Market price changes based on supply/demand. You define a max price.
- **Interruption:** If Spot Price > Max Price, or if AWS needs capacity back, you get a **2-minute warning**.
- **Spot Request Types:**
  - *One-time:* If interrupted, the request is closed.
  - *Persistent:* If interrupted, the request opens again to try and launch a new instance when capacity is available.
- **Spot Fleet:** A collection of Spot Instances and (optionally) On-Demand Instances.
  - It attempts to meet a target capacity (e.g., 50 vCPUs) using a mix of instance types.
  - **Strategies:**
    - **lowestPrice:** Launches from the pool with the lowest price (Cost optimization, but higher interruption risk).
    - **diversified:** Distributed across all pools (High availability, lower interruption risk).

        - **capacityOptimized:** Launches from pools with optimal capacity for the number of instances (Best for minimizing interruptions). *Exam Favorite.*
- *Exam Tip:* If a question asks for "Cost optimization" for a "Critical workload" or "Database," **NEVER** choose Spot Instances. Spot is only for **interruptible** workloads.